

RESEARCH CASE STUDY

AUM SHINRIKYO

AND THE AI UPLIFT PROBLEM

HOW LLMS COULD LOWER THE THRESHOLD FOR CBRN HARM



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How LLMs Could Lower the Threshold for CBRN Harm

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Executive Summary

Aum Shinrikyo remains one of the most important historical cases for understanding violent non-state interest in CBRN capabilities. The group had apocalyptic intent, substantial financial resources, educated personnel, international reach, and a demonstrated willingness to cause mass harm. Those advantages still did not produce a reliable capability. Again and again, Aum struggled to turn ambition into execution.

That gap makes the case useful for frontier AI safety. Large language models do not have to create dangerous capabilities from scratch to change the equation. They can help users organize scattered information, impose structure on vague goals, and keep complex work moving. Many of Aum's weaknesses lie as much in cognitive and organizational spaces as in technical ones.

This paper argues that frontier LLMs can create threshold-lowering risk for motivated non-state actors. Ceiling-raising risk concerns highly capable actors using advanced AI systems to pursue more optimized, scalable, novel, or catastrophic capabilities. Threshold-lowering risk looks different. It concerns actors who already have intent, resources, and partial expertise but still struggle to cross the practical threshold into credible harm. Aum fits that category. It was neither a lone novice nor a state weapons program. It sat in the dangerous middle ground where better planning, interpretation, coordination, and learning could have changed the outcome.

LLMs would not have guaranteed Aum's success. They would not have removed the physical constraints, safety hazards, tacit knowledge gaps, or organizational dysfunction that shaped the group's failures. The narrower concern is that LLMs can remove enough friction across a harmful workflow to change an actor's risk profile. In CBRN contexts, violent actors do not need technical perfection to create fear, disruption, casualties, or strategic attention.

For frontier AI labs, the implication is direct. Safety evaluations should examine cumulative uplift across longer human-AI workflows, especially for users ranging between novices and experts. A single response may look harmless in isolation and still contribute to risk when it supports a sustained effort.

This paper uses Aum Shinrikyo as a historical stress test for that problem. It does not provide operational CBRN details or reconstruct technical pathways for harm. It examines where Aum failed and asks how frontier LLMs could affect those failure nodes. Aum does not show that AI would make CBRN harm easy. It shows something narrower and more uncomfortable: a future actor may only need AI to become organized, adaptive, and competent enough to keep moving.

About the Author

Matthew Tripoli is a U.S. Navy Explosive Ordnance Disposal Chief Petty Officer and Master EOD Technician with more than 15 years of experience in CBRNE threat assessment, explosive hazards, hazardous materials response, and special operations support. He served as a high-risk instructor in the Chemical and Biological Weapons Division at the Naval School Explosive Ordnance Disposal, where he is a Master Training Specialist in weapons recognition, threat analysis, and operational response to joint and allied military personnel. His current work focuses on the intersection of CBRN threats, misuse of artificial intelligence, and national security risks. He holds a Master of Arts in Homeland Security with a focus on Weapons of Mass Destruction and Risk Management.

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AI Assistance Disclosure

The author used tools such as Grammarly and large language models during the development of this case study to outline, edit, organize, create images, and conduct a critical review. All substantive claims, historical details, and source-based analysis were the creation of the author and supported through cited materials. The author retains responsibility for the final argument and accuracy of the work.

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Introduction

Aum Shinrikyo remains one of the most important cases for understanding non-state interest in Chemical, Biological, Radiological, and Nuclear (CBRN) capabilities. The group combined apocalyptic ideology, substantial financial resources, educated personnel, and a willingness to pursue mass-casualty violence (Lifton, 2005). Those advantages did not translate cleanly into reliable capability. Aum's history exposes the gap between ambition and execution.

That gap now has direct relevance to frontier AI safety. LLM risk does not rest solely on whether a model provides an isolated piece of dangerous information. Much of that information already exists in fragmented public or semi-public sources. The harder issue is usability: whether a model helps a motivated user synthesize, sequence, troubleshoot, and organize complex activity.

This case study asks how frontier LLMs might reduce the barriers between violent intent and effective CBRN execution for an Aum-like actor. LLMs would not erase the real-world constraints that shaped Aum's failures. They could still change the risk profile by helping motivated users organize knowledge, troubleshoot setbacks, and learn faster (Pannu et al., 2025).

This analysis uses intentional boundaries for both methodological and safety reasons. It focuses on LLMs rather than the full range of biological AI systems, and it avoids operational CBRN detail. This narrower scope makes Aum Shinrikyo a useful historical stress test of how AI affects the gap between violent intent and practical execution.

Scope and Method

Technical Scope

This paper focuses on large language models rather than biological design tools, autonomous laboratories, protein design systems, or other specialized scientific AI systems. LLMs are widely accessible and help users reason, plan, synthesize, and perform above their baseline (Wheeler, 2025). Although they do not replace laboratory infrastructure or hands-on technical skill, they can still change how users approach complex problems.

Actor Scope

The study focuses on a motivated non-state actor with intent, resources, and some technical access, but limited ability to convert those inputs into a reliable capability. Aum Shinrikyo fits that purpose because it had unusual motivation and resources, yet still struggled with execution, technical integration, and organizational learning (Bleek, 2011). Aum was an extreme case, not a representative one. Its value lies in the combination: the group had enough capacity to matter, but enough failure to show where its ambition broke down.

Definition of Uplift

This paper uses "uplift" to mean a measurable or plausible improvement in an actor's ability to move from intent toward execution. Uplift does not require AI to create a capability from nothing, replace domain experts, or guarantee success. It can occur when a model turns partial understanding into more coherent action. Users experience uplift when they achieve more with the same resources or reach a similar outcome with fewer resources (Rose et al., 2024).

Method

This paper uses a structured historical case analysis. It examines Aum through five categories of actor capability.

Intent: The group's motivation and willingness to commit mass harm.

Access: The group's ability to obtain money, people, facilities, information, and other resources.

Technical Capability: The group's ability to turn those resources into a functioning, harmful capability.

Execution: The group's ability to coordinate, test, adapt, and act under real-world constraints.

Learning: The group's ability to absorb failure, correct mistakes, and improve over time.

The analysis then asks how LLMs could affect each category without assuming that AI removes physical, organizational, or security constraints. This approach keeps the paper focused on changes in actor capability rather than isolated pieces of information.

Safety Boundaries

This paper avoids operational CBRN detail. It does not provide instructions, technical parameters, acquisition guidance, design specifications, or procedural pathways that could cause harm. It discusses historical examples only at the level needed to understand organizational barriers and capability limits. That boundary keeps the paper focused on AI safety risk rather than making dangerous knowledge more usable.

Limits

This analysis has several limitations. It does not test live model outputs, evaluate every AI system relevant to biosecurity, or assess specific frontier AI companies or individual models. It also does not argue that LLMs would have made Aum successful or that current safeguards fail across the board. The narrower point is that LLMs may lower cognitive and organizational barriers that historically limited motivated actors. Those marginal changes can matter even when they do not guarantee operational success.

Threshold-Lowering vs. Ceiling-Raising Risk

Analysts often discuss AI-enabled CBRN risk as if it were one problem. It is more useful to separate the risk into two problems. Threshold-lowering expands credible capability among motivated actors who previously struggled to execute. Ceiling-raising expands the maximum capability of actors who already possess advanced expertise, infrastructure, and resources. Aum Shinrikyo is a better stress test for the first problem.

Threshold-Lowering Risk

Threshold-lowering risk occurs when AI helps a motivated actor reach a level of capability that previously sat beyond its practical reach. The actor does not need a novel threat or the worst possible outcome. It only needs enough capability to create operationally relevant harm. Aum fits this profile. It lacked neither violent intent nor money nor educated members. Its central problem involved converting those inputs into a reliable capability (Quillen, 2025). That makes the case useful for studying the space between ambition and execution.

Many violent actors do not need technical perfection. A crude or unreliable capability can still cause serious effects if it reaches a target, creates public fear, overwhelms responders, or signals that a group has crossed a taboo (Sugiyama et al., 2020). For AI safety, the relevant question is whether LLMs help motivated actors become sufficiently capable of causing credible harm.

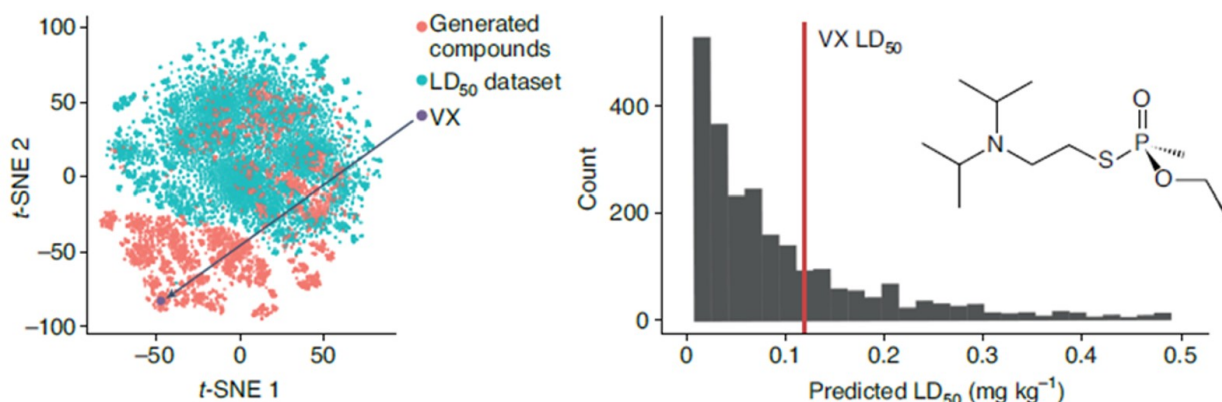
Ceiling-Raising Risk

Ceiling-raising risk refers to the possibility that advanced AI helps already capable actors pursue more optimized, scalable, novel, or catastrophic capabilities. This risk profile applies more directly to states, state-backed programs, advanced laboratories, or unusually sophisticated non-state actors. These actors may already have personnel, infrastructure, and technical depth. For them, AI may expand the upper bound of what they can design, test, or optimize.

Urbina et al. (2022) illustrate this ceiling-raising concern in the chemical domain. In their proof-of-concept study, researchers inverted a drug-discovery model that normally helps identify therapeutic molecules and instead directed it toward toxicity. The system generated thousands of candidate molecules in a short period, including known chemical warfare agents and novel molecules predicted to be highly toxic. The authors did not synthesize the molecules and withheld operational details, but the experiment showed how a beneficial discovery tool could become dangerous when users reverse its objective function. The concern centers on advanced tools expanding the design space available to actors who already possess relevant technical capacity, infrastructure, or access to specialized workflows.

Figure 1

A t-SNE plot visualization of the LD₅₀ dataset and top 2,000 MegaSyn AI-generated and predicted toxic molecules illustrating VX.



Note. Reproduced from *Dual use of artificial-intelligence-powered drug discovery*, by Urbina et al., 2022, p. 189. "Many of the molecules generated are predicted to be more toxic in vivo in the animal model than VX (histogram at right shows cut-off for VX LD₅₀). The 2D chemical structure of VX is shown on the right."

Ceiling-raising risk remains important, but it is not the main focus here. That risk often involves specialized biological AI systems, laboratory automation, advanced scientific workflows, and technical expertise beyond the scope of an LLM-focused case study. Separating it from threshold-lowering risk allows the Aum case to examine the problem it best illuminates.

Why the Distinction Matters

Different risks require different evaluations. A ceiling-raising evaluation asks whether AI enables the most capable actors to do something more advanced than they could otherwise. A threshold-lowering evaluation asks whether AI helps less capable but motivated actors cross a practical line into credible harm.

Combining these risks can push policymakers and frontier labs toward the wrong safety questions (Anderljung, Hazell, & von Knebel, 2024). Evaluations should look beyond whether a model produces extreme or highly technical information. They should also test whether a model improves a user's ability to organize, plan, troubleshoot, and iterate across a multi-step project (Spelda & Stritecky, 2025). The most important uplift may appear as a sequence of ordinary interactions that, over time, make a motivated user more capable.

Aum Shinrikyo as a Historical Case

Motivation and Ideology

Aum's interest in CBRN capabilities cannot be separated from its ideology. The group began as a self-help organization led by Chizuo Matsumoto, who later changed his name to Shoko Asahara. Over time, Aum adopted a more apocalyptic and violent worldview. Its leadership built a moral framework that allowed members to justify mass harm, and even treat it as necessary, in the service of a larger mission to trigger the apocalypse. Most violent groups do not pursue CBRN capabilities, even when those capabilities might generate attention. Aum's worldview helped push the group across that boundary. Opportunity alone was not enough; members needed a moral framework that made taboo violence seem permissible, even necessary (Asal, Avdan, & Ackerman, 2023).

From its inception in 1987 to 1990, Aum quickly moved beyond ideological interest into the active pursuit of CBRN weapons. The group repeatedly tried to turn that interest into usable capability. Before turning to chemical weapons, they focused heavily on biological weapons. Lacking access to the Ebola virus or the practical knowledge to cultivate it, they sent personnel to the Congo in a failed attempt to retrieve a strain in 1992 (Tu, 2014). They also pursued anthrax and the botulinum toxin.

Aum's pursuit of *C. botulinum* reflected the group's ideological preference for self-reliance. Although Aum operated a hospital and may have had plausible ways to obtain small amounts through research channels, leadership decided to attempt to collect from the soil in Japan's Ishikarigawa Basin (Danzig et al., 2012). This choice fits a broader pattern in which the group tried to perform tasks internally, even when outside acquisition may have offered a faster or more efficient path. That preference likely reflected some combination of secrecy, devotion to Asahara, and belief that self-produced materials were cheaper, faster, or more spiritually consistent with the group's worldview (Danzig et al., 2012). Ideology shaped both the group's willingness to pursue mass-casualty weapons and the inefficient pathways it chose.

Resources and Human Capital

Aum also had resources that most violent non-state actors lack. It had significant financial assets, access to facilities, international reach, and a pool of educated members. Its membership included individuals with scientific, technical, and medical training (Tanimoto et al., 2018). Seiichi Endo, a graduate student in virology and genetic engineering, ran Aum's biological weapons program. Masami Tsuchiya, a chemist, and Dr. Tomomasa Nakagawa, Asahara's personal doctor, produced the group's first batch of sarin.

That placed Aum between the low-capability lone actor and the state-sponsored weapons program. This middle ground makes Aum especially relevant to threshold-lowering risk. The group did not start from scratch, but it also lacked the disciplined expertise, tacit knowledge, and technical integration of a state weapons program. Its trained members could pursue complex projects, but their credentials did not translate into reliable execution (Nehorayoff, Ash, & Smith, 2016). That is the danger zone: people, resources, and partial technical access without the discipline needed to use them well.

Ambition and Technical Failure

Aum's CBRN ambitions were broad and serious. The group pursued multiple pathways and repeatedly tried to move from aspiration to operational capability. It ultimately pursued several chemical agents, including sarin (GB), soman (GD), tabun (GA), cyclosarin (GF), VX, mustard agent, hydrogen cyanide (AC), and phosgene, a chemical with industrial uses that also functions as a choking agent (Tu, 2014). However, Aum's results remained uneven.

Satyan 7, one of Aum's production facilities, illustrates this gap clearly. The facility supported the group's effort to produce sarin on a large scale, with reported ambitions to manufacture tens of tons of the agent. However, Aum's chemists had gained early success by starting later in the production process with a purchased precursor. They had not mastered the earliest stage of production in their own laboratory. When Asahara pushed for rapid scale-up, the group failed to produce the necessary precursor internally, and the process reportedly led to chemical leaks from the facility in July 1994 (Tu, 2014). A single missing ingredient did not

cause that gap. The failure reflected technical limits, weak project discipline, poor judgment, organizational secrecy, and insufficient practical knowledge (Bleek, 2011).

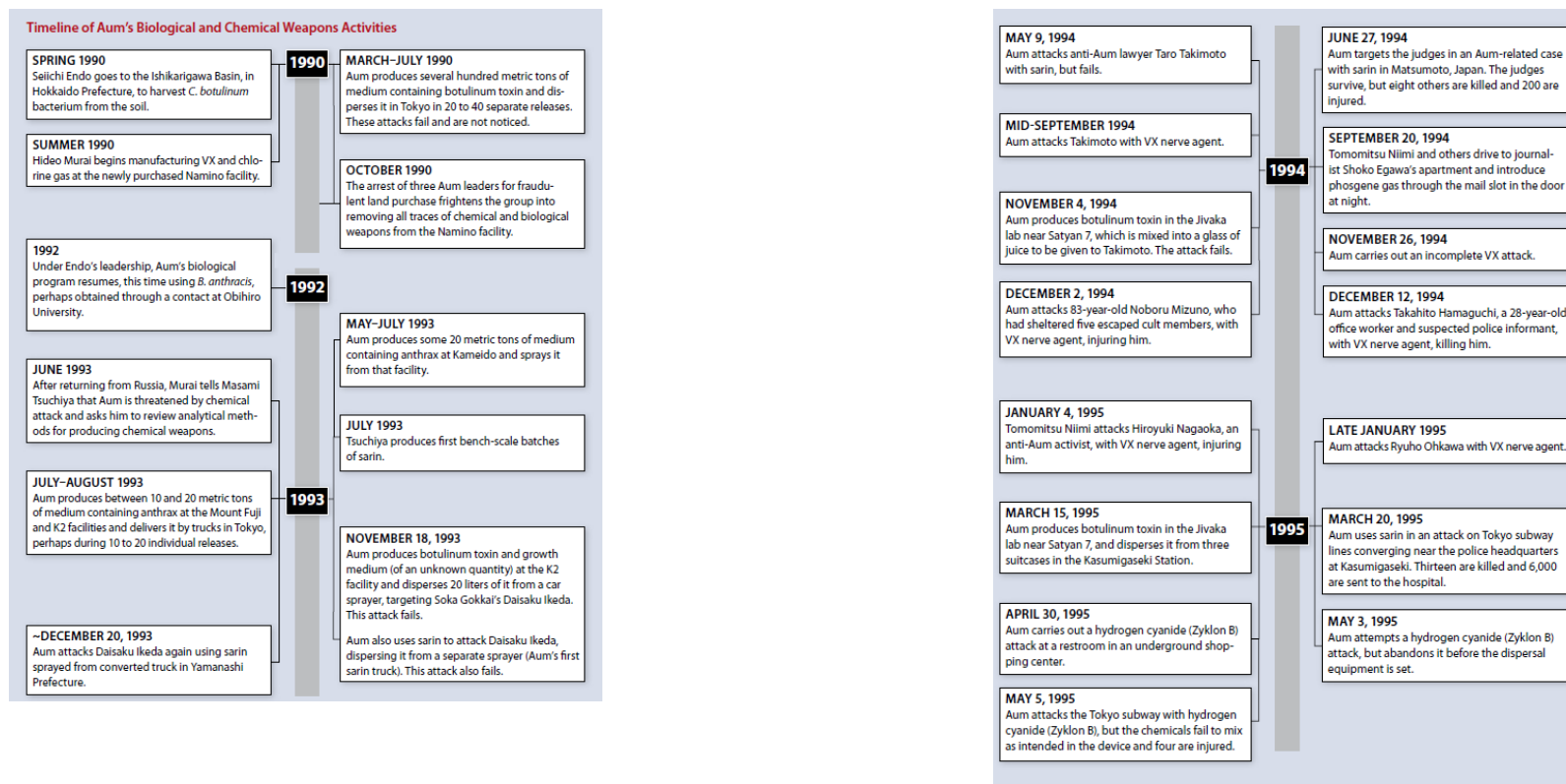
Aum's botulinum program showed the same pattern on the biological side. The group reportedly produced very large quantities of liquid material associated with its *C. botulinum* effort, but the volume did not translate into an effective weapon. The material appears to have consisted mostly of growth media rather than concentrated or purified toxin. Even so, Aum attempted to use it operationally. The group reportedly used three trucks to spray the material at several high-profile targets, including two U.S. naval bases. When these attacks failed, Aum leaders blamed several possible causes, including a weak pathogen, malfunctioning spray equipment, and poor wind prediction (Danzig et al., 2012). Internal observations reinforced the failure. One member fell into a fermentation tank and showed no signs of illness, while later animal testing produced results that did not clearly confirm toxicity (Danzig et al., 2012).

The anthrax program followed a similar pattern. Aum's *B. anthracis* program appears to have failed for reasons similar to the botulinum effort. Aum likely did not produce a fully virulent strain, and even if their scientists understood what they needed to achieve, they lacked the practical skill and technical control to make it work. The process created several failure points before Aum reached operational use (Danzig et al., 2012). The cult's own members reportedly began to doubt the potency of the material after accidental exposures and failed attacks produced no illness.

Aum repeated a familiar mistake. It moved into large-scale production without confirming that the material could cause the intended effect. The anthrax preparation reportedly consisted of an unpurified slurry dominated by growth media, producing a foul-smelling material that drew complaints from nearby residents and police attention. The group tried to overcome dissemination problems through improvised equipment, but the spraying systems repeatedly malfunctioned. Instead of solving the underlying biological and engineering problems, Aum continued to improvise around them. Aum's members could produce bulk material, build devices, and conduct attacks, but they could not reliably convert those actions into an effective biological weapon.

Figure 2

Timeline of Aum's Biological and Chemical Weapons Activities



Note. Reproduced from *Aum Shinrikyo: Insights Into How Terrorists Develop Biological and Chemical Weapons*, by Danzig et al., 2012, p. 20-21.

Organizational Dysfunction

Aum's limitations extended beyond the technical work. Its internal structure damaged judgment and execution. Asahara concentrated authority around himself, while ideology, fear, secrecy, and magical thinking distorted practical decision-making (Lifton, 2005). He also reportedly used multiple members for criminal tasks, creating a system in which offenders monitored one another while responsibility diffused across the group. Instead of increasing accountability, this arrangement made dissent less likely. Each participant could assume that if an order had truly crossed a line, someone else would object. This structure helped members participate in serious crimes without experiencing the decision as fully individual (Danzig et al., 2012).

The Sakamoto murders, one of the group's early crimes, revealed patterns that later appeared in Aum's biological and chemical attacks: shifting methods, senior-member execution, compartmentalized knowledge, and crude but effective violence (Danzig et al., 2012). These features reflected a dangerous organizational mix. Aum could act decisively when Asahara gave direction, but it lacked the disciplined feedback and decision controls needed for complex technical projects. The result was not simple incompetence. The group could cause serious harm despite poor planning and unstable internal judgment.

This dysfunction also appeared in the group's handling of sarin. After Asahara ordered his scientists to destroy their sarin stockpile before authorities could discover it, they reportedly had such a large quantity that they could not complete the task. After three days of work while self-administering 2-PAM chloride, a nerve agent antidote used to counter acetylcholinesterase inhibition, they gave up and hid the dangerous sarin precursors instead (Tu, 2014). The episode shows how scale, secrecy, and poor control turned Aum's own resources into management problems.

Harm from Imperfect Execution

Aum's history shows that imperfect execution can still cause serious harm. The 1995 Tokyo subway attack fell short of the group's imagined maximum, but it still produced deaths, injuries, public fear, and long-term health consequences for survivors (Miyaki et al., 2005). Aum already possessed a substantial supply of the sarin precursor difluoromethylphosphonic acid, which allowed the group to move quickly into final production before the attack. Time pressure pushed Aum to use an unpurified, finished agent. A purified product would have made the attack much more dangerous (Tu, 2014).

The attack shows that a capability need not function perfectly to cause strategic harm. Aum was dangerous despite its failures. It had enough intent, enough material, and enough capacity for execution to act, even while its broader program remained uneven. That combination makes the case useful: Aum was dangerous enough to matter, but flawed enough to reveal where friction constrained capability.

Analytical Bridge: Aum Failure Nodes and AI-Relevant Friction

Table 1
Aum Shinrikyo Failure Node Map

Aum Failure Node Map AI-relevant friction points

Failure node	Historical anchor	Friction and LLM relevance	Remaining constraint
Intent-to-Execution Gap Intent / Execution	Aum had extreme intent, significant financial resources, educated personnel, and infrastructure, yet it repeatedly struggled to translate ambition into a reliable CBRN capability.	Friction: Ambition did not translate cleanly into capability. LLM relevance: LLMs could help structure the path through organized project planning, technical literacy, and staged decision-making.	Physical execution is still required.
Biological Agent Selection and Acquisition Failure Access / Technical Capability	Aum struggled to obtain or work with effective biological agents.	Friction: Users may misunderstand what they have, what they need, or why a biological effort is failing. LLM relevance: An LLM could help a user understand categories, constraints, terminology, and invalid assumptions without operational guidance.	Physical execution still required; safeguards should block actionable acquisition guidance.
Tacit Knowledge Gap Technical Capability	Aum had some relevant expertise but lacked specialized tacit knowledge to isolate, cultivate, manipulate, and sustain organisms.	Friction: Formal education did not equal practical competence. Tacit knowledge is often the difference between reading about a process and making it work. LLM relevance: LLMs could partially compensate by explaining concepts, sequencing learning, and helping non-specialists recognize gaps in their understanding.	Lab judgment, safe handling, contamination control, and validation still require real experience.
Crude Biological Production Approach Technical Capability	Aum's biological production efforts were crude, contaminated, and ineffective, including efforts that failed to produce casualties.	Friction: Aum struggled to diagnose why production efforts were failing. LLM relevance: LLMs could improve high-level troubleshooting logic by identifying assumptions to check, broad failure categories, and the evidence needed to distinguish them.	Models should not provide procedural optimization or harmful operational details. Physical validation remains unavoidable.
Ineffective Biological Dissemination Execution	Even if Aum had obtained potent agents, dissemination remained a major secondary hurdle, and crude techniques likely limited casualties.	Friction: Aum underestimated the complexities of environmental, delivery, exposure, and self-protection. LLM relevance: LLMs could help actors reason about feasibility, constraints, and failure points at a high level. Risk may also arise when users seek guidance from adjacent domains rather than explicit CBRN intent.	Safe models must refuse actionable dissemination assistance and recognize adjacent-domain requests that would create CBRN-relevant operational guidance. Real-world testing, exposure risk, and detection risk remain major barriers.
Trial-and-Error Persistence Learning	Aum kept trying despite repeated failures.	Friction: Repeated failure did not stop the group. It created learning opportunities. LLM relevance: LLMs could shorten learning cycles by helping users compare failed attempts, organize lessons learned, and avoid repeating obvious mistakes.	LLMs cannot guarantee a correct diagnosis and may hallucinate. Safeguards may interrupt risky patterns. Detection and material constraints persist.
Weak Decision Discipline Execution / Learning	Asahara's authority distorted planning and reality testing.	Friction: Bad decisions were not just technical. They were organizational and overrode disciplined evaluation. LLM relevance: LLMs could provide structured reasoning, assumption tracking, option comparison, checklists, and critique of flawed plans.	AI cannot remove cult authority, delusion, coercion, fear, or leader dominance. Leaders can always ignore good advice.
Overreach Across Advanced Technologies Learning / Execution	Aum pursued nuclear, chemical, biological, laser, plasma, seismological, and other exotic ideas, many closer to fantasy than feasible operations.	Friction: The group lacked disciplined feasibility screening. LLM relevance: LLMs could help rank ideas by feasibility, uncertainty, resource burden, and likely bottlenecks, potentially steering actors away from fantasy projects and toward more feasible harmful pathways.	Safe systems should avoid feasibility-ranking support for harmful projects. Real-world access, expertise, cost, and detection risk still constrain execution.
Personnel Mismatch Access / Technical Capability	Aum had scientists and engineers, but not always the right expertise for the specific problem at hand.	Friction: Credentials did not equal relevant capability. LLM relevance: LLMs could help partially trained personnel translate adjacent expertise into more useful contributions to projects.	AI cannot turn an untrained person into a true specialist or replace validated laboratory experience.
Project Coordination Across Ministries / Units Execution	Aum had ministries and internal units, but complex projects still suffered from poor integration, secrecy, and distorted reporting.	Friction: Fragmented teams lost feedback, hid bad news, and repeated mistakes. LLM relevance: LLMs could improve project management, task decomposition, documentation, timeline tracking, and dependency mapping.	LLMs cannot solve mistrust, compartmentalization, or fear-driven reporting.
Harm Threshold Below Technical Perfection Execution	Aum's chemical program caused harm even though execution was far from optimal; the Tokyo attack caused mass injuries but fell short of the intended results.	Friction: The threshold for harm is lower than the threshold for technical perfection. LLM relevance: LLMs could increase risk by helping a group become "good enough" rather than an expert. Marginal improvement may push actors across a practical threshold.	Physical handling, deployment risk, detection, and poor judgment remain major constraints.

Note. Adapted from *Revisiting Aum Shinrikyo: New Insights into the Most Extensive Non-State Biological Weapons Program to Date* (Bleek, 2011), *Aum Shinrikyo's Biological Weapons Program: Why Did it Fail?* (Rosenau, 2001), *Aum Shinrikyo's Nuclear and Chemical Weapons Development Efforts* (Nehorayoff, A. A., Ash, B., & Smith, D. S., 2016).

The table above maps Aum's recurring failure nodes to the five capability categories used in this analysis: intent, access, technical capability, execution, and learning. These categories overlap in practice. Access problems affect technical capability. Technical failures affect execution. Repeated failure can stop a project or feed a learning cycle. The table uses the five-category framework to identify where Aum struggled and where LLMs could plausibly reduce friction. Taken together, these failure nodes explain why Aum is more than a historical CBRN case. It gives AI safety researchers a way to examine how planning, interpretation, coordination, and learning can break down.

LLM Uplift Pathways

This section examines how LLMs affect the failure nodes identified in the Aum case. These pathways operate below the level of direct operational instruction. They rely on ordinary model capabilities: explaining, comparing, planning, critiquing, translating, troubleshooting, and remembering. Many of Aum's failures were organizational and interpretive rather than purely informational. That places ordinary LLM capabilities directly inside the problem space.

Recent AI-biosecurity work supports this capability-based frame. The Centre for Long-Term Resilience (CLTR) defines uplift as an improvement in what an actor can accomplish with a given level of resources, or the ability to reach a similar outcome with fewer resources. It also identifies several relevant consequences of uplift, including faster development, reduced expertise requirements, and a higher likelihood of success (Rose et al., 2024). Pannu et al. similarly argue that biosecurity evaluation should focus on capabilities of concern rather than only on specific pathogens or fixed threat scenarios. That approach fits the Aum case because the relevant issue is whether an LLM could reduce enough friction to make an already motivated actor more capable.

The subsections below examine seven uplift pathways: knowledge compression, troubleshooting and failure interpretation, planning and project management, decision discipline, coordination and memory, expertise barrier lowering, and speed and iteration. Each pathway has legitimate, benign uses, which makes the safety problem difficult to address.

Knowledge Compression

LLMs compress scattered research into usable explanations. Aum did not operate in a total information vacuum. It had educated members, outside publications, internal specialists, facilities, and time. Its problem involved interpretation: members had to identify what mattered and connect information across domains.

Aum's biological program shows the difference between information access and usable knowledge. Rosenau argues that Aum possessed many inputs associated with a serious program, including funding, dedicated personnel, facilities, and scientific training, yet still failed to develop a mass-casualty biological capability. The group struggled with effective agent acquisition, scientific judgment, dissemination, and organizational conditions that distorted decision-making. Tu's account reinforces the same pattern. Aum initially emphasized biological weapons, but the program failed, while the chemical program achieved more operational success under personnel whose expertise better matched the task. In both accounts, the problem was not simply the absence of information. It was the failure to convert partial information and partial expertise into a reliable capability.

LLMs could affect that barrier by making relevant knowledge more usable. A model can explain terminology, summarize long sources, and help users recognize when their background does not match the problem they face. The greatest marginal effect may come from helping technically literate users recognize what they do not understand.

Compression can also mislead. A model may oversimplify, omit critical caveats, or create a false sense of confidence by making a complex subject appear more achievable than it really is. Usable summaries do not replace practical judgment. Even a partial reduction in confusion could help a motivated actor move through the early stages of understanding more quickly than they would otherwise.

Troubleshooting and Failure Interpretation

Failure does not automatically produce learning. A group must interpret what went wrong before it can correct course. Aum repeatedly encountered failed or ineffective outcomes, especially in its biological program, but it lacked the scientific judgment and testing discipline needed to turn those failures into reliable corrections. Rosenau identifies several reasons for the biological program's failure, including poor scientific work, poor fit between expertise and the task, ineffective dissemination, and organizational conditions that distorted judgment (Rosenau, 2001). The group did more than fail. It often failed without understanding why.

Many harmful projects fail not because they are impossible, but because actors misread the source of failure. Without disciplined failure interpretation, repeated attempts can reproduce the same mistakes. Aum showed that pattern. It kept trying, but its learning remained uneven because the organization lacked the practical expertise and reality testing needed to diagnose failure correctly.

LLMs could strengthen the interpretation stage. A model can help users sort possible explanations and organize a failure review. It can also help distinguish between high-level categories of failure, such as flawed assumptions, poor execution, missing expertise, or inadequate validation. The CLTR report identifies practical laboratory troubleshooting tasks as a potential source of uplift, as AI systems may help users fill gaps left by formal protocols and provide personalized, iterative assistance when tasks do not work as intended. The report also links troubleshooting to tacit knowledge barriers: the informal know-how that practitioners often gain through experience rather than written instructions (Rose et al., 2024).

Troubleshooting uplift does not require a perfect diagnosis. Even imperfect feedback may shorten the distance between one failed attempt and a more informed next attempt. In Aum's case, failure was not a stopping point. It became part of the group's learning cycle.

Planning and Project Management

Aum also struggled with project structure. The group pursued complex efforts that required sequencing, feedback, and control across stages. Its hierarchy helped initiate and resource chemical and biological programs, but it also concentrated control in leaders who often made poor judgments about technical feasibility and project execution. Satyan 7 illustrates this problem. Aum built an expensive, technically ambitious facility, but it failed to transition from laboratory-scale chemistry to reliable mass production (Danzig et al., 2012).

LLMs could affect this barrier by improving project structure. A model can turn a vague objective into a structured plan with milestones, dependencies, and open questions. It can help users keep work aligned over time, identify missing prerequisites, and organize information in a way that makes a complex effort easier to manage. This kind of support does not look like direct technical instruction, but it can still change actor capability.

RAND's AI Implications for CBRN framework reinforces this concern by emphasizing pathway analysis from ideation through capability development, production, weaponization, and operational deployment. That framing treats planning support as part of the risk pathway rather than a neutral background function. If risk develops through stages, then assistance that helps a user understand, organize, or move between stages can contribute to uplift even without providing operational details (Hersman, Blackwelder, & Parachini, 2026). Project management uplift does not require perfect accuracy. It only requires enough structure to help a motivated actor use existing resources more efficiently.

Decision Discipline

Decision discipline deserves separate treatment because it differs from project management. Project management asks whether a group can execute a plan. Decision discipline asks whether the group chooses a sensible plan in the first place. For violent non-state actors, this distinction matters. A group may have intent, resources, and personnel, yet still waste time on pathways that do not align with its actual capabilities.

Aum often failed this test. The group spread attention across chemical, biological, nuclear, seismological, plasma, laser, and exotic weapons concepts. Some of these ideas reflected ideology, status competition, science-fiction influence, and Asahara's fascination with technologically advanced destruction more than disciplined feasibility analysis. Nehorayoff, Ash, and Smith argue that Aum's engineering ambitions repeatedly collided with unrealistic assumptions, ideological pressure, and unstable leadership judgment (Nehorayoff, Ash, & Smith, 2016). Danzig et al. similarly describe Aum's weapons development as erratic rather than methodical, with resources and attention often loosely tied to technical merit (Danzig et al., 2012).

A group that wastes resources on fantasy projects may remain dangerous, but it will likely be less effective than a group that can separate feasible pathways from delusion. Aum's failures included poor execution and poor selection. The group sometimes chose projects because they fit apocalyptic imagination, internal status incentives, or the leader's preference, rather than because they offered the most realistic route from intent to harm.

LLMs could affect this failure mode by sharpening feasibility analysis. A model can help users compare options, surface assumptions, and clarify what evidence would change a decision. It can also help translate vague goals into clearer decision criteria. This kind of support can improve how a user thinks about goals, tradeoffs, and feasibility.

LLMs would not have cured Aum's deeper organizational pathologies. Leaders can ignore good advice, and models can produce confident but flawed analysis. Even so, imperfect decision support could still change how a motivated actor allocates attention.

Coordination and Memory

Aum's secrecy created a tradeoff. It helped conceal sensitive activity from outsiders, but it also narrowed access to expertise, reduced feedback, and weakened coordination. Danzig et al. argue that Aum's key biological and chemical work largely stayed within the leadership group. This compartmentalization made secrecy easier, but it also limited the skill sets available for weapons development and forced the group to rely on trusted insiders rather than the most qualified personnel (Danzig et al., 2012).

That tradeoff shaped Aum's failures. The group often favored internal workarounds over outside acquisition, even when external materials, services, or expertise might have solved problems more efficiently. This pattern reflected secrecy, self-reliance, and loyalty to Asahara, but it also created practical costs. Aum's internal structure made it harder to obtain a dispassionate technical assessment, harder to identify why projects failed, and easier for bad assumptions to persist. When a problem arose, the group could not easily seek outside review without increasing exposure risk. As a result, secrecy did not merely hide the program. It also degraded the program's ability to learn.

LLMs could reduce some of these coordination costs. A model can preserve decisions, track open questions, and translate technical issues into terms non-specialists can understand. The CLTR report's definition of uplift aligns with this pathway because coordination and memory can help an actor accomplish more with the same resources (Rose et al., 2024).

Expertise Barrier Lowering

Aum's personnel problem was not a simple lack of intelligence or education. The group included people with scientific, technical, and medical backgrounds. The problem was fit. The personnel available to Aum did not always match the specific technical demands of the projects they pursued. Rosenau argues that Aum had the motivation, resources, and persistence but lacked the full range of scientific and technical expertise required for an effective biological weapons program (Rosenau, 2001). The gap included not only biological knowledge but also practical skills in engineering, environmental conditions, and dissemination. Tu's account reinforces the same point: Aum's biological program suffered because Endo lacked the practical background needed for the biological agents he attempted to develop, whereas the group later achieved greater success under Tsuchiya's chemical expertise (Tu, 2014).

Expertise does not operate as a simple binary. Aum's members could read technical material, build facilities, acquire equipment, and attempt production. They still struggled to convert adjacent knowledge into reliable capability.

LLMs could lower this barrier by making adjacent expertise more useful. A model can help a technically literate member understand unfamiliar terminology, frame better questions, and translate between domains. The tool would not make that person a specialist, but it could help them use partial knowledge more effectively.

Quillen's work on terrorist chemical and biological weapons (CBW) selection strengthens this point. He argues that terrorist leaders often authorize CBW efforts, while scientific and technical personnel make key choices about which agents to pursue and how to pursue them (Quillen, 2025). The relevant uplift target, then, includes the technical intermediary: the educated insider trying to turn leadership intent into practical work. If LLMs help that person operate across knowledge gaps, they may expand the number of personnel who can meaningfully support a harmful project.

Speed and Iteration

Speed changes risk because persistent actors improve through repeated cycles. Aum's history shows that failure did not necessarily terminate a dangerous project. The group continued pursuing chemical and biological capabilities despite accidents, failed attempts, ineffective materials, equipment problems, and bizarre technical ideas. Danzig et al. treat this persistence as one of the central lessons of the case: repeated failure can reveal weakness, but it can also signal that a motivated group continues to search for a workable path (Danzig et al., 2012). Failure should not automatically reassure analysts. With a committed actor, each failed attempt can create another opportunity to adjust assumptions, change methods, or redirect effort.

Aum's movement across biological and chemical programs illustrates this dynamic. The group shifted among biological and chemical pathways, returning to earlier efforts after failures, and continued to modify crude approaches. These changes did not reflect disciplined research and development in the state-program sense. They reflected persistence, improvisation, and uneven learning under ideological pressure. Partial capability was enough to make the group dangerous, and repeated attempts gave it more chances to find workable paths.

LLMs could accelerate this learning cycle. They can help users summarize what happened, organize lessons learned, and prepare for the next attempt more quickly. They can also maintain continuity across sessions by preserving task history, decisions, constraints, and prior mistakes. These functions do not guarantee correct diagnoses or successful outcomes. They can still reduce the time and effort required to move from failure to renewed action. A group that learns slowly poses a particular risk. A group that can review, reorganize, and iterate faster poses another challenge.

This safety problem does not fit neatly into single-turn refusal models. The most important assistance may never appear as one explicit request for prohibited information. It may appear through ordinary interactions: summarizing prior work, comparing options, organizing a timeline, identifying bottlenecks, or helping a user decide what to revisit next. Spelda and Stritecky's critique of AI security practices is relevant here because they argue that alignment and red teaming cannot provide dependable safety guarantees across the full range of future model behavior (Spelda & Stritecky, 2025). Testing can reduce risk, but it cannot cover every context in which a general-purpose model may, over time, support harmful activity.

For threshold-lowering risk, speed and iteration compress the distance between intent, failure, learning, and renewed execution. Aum's failures slowed, disrupted, and sometimes redirected the group. They did not reliably stop it. If LLMs help future actors preserve lessons, interpret setbacks, and iterate more efficiently, they may change an actor's risk profile without providing direct operational instructions. The concern is not instant success. It is a faster movement through the learning cycle.

Limits and Counterarguments

This argument has important limits. Aum Shinrikyo helps examine how LLMs could lower some barriers between violent intent and practical execution. The case still does not prove that AI would make an Aum-like actor successful. The more defensible claim is that LLMs may change how some constraints operate by making knowledge easier to use, planning easier to organize, and failure easier to interpret. Those changes can alter the risk profile without erasing the real-world barriers that shape CBRN capability.

The first counterargument is that information is not operational capability. CBRN activity requires more than explanations on a screen. Aum had significant advantages, yet those inputs did not automatically translate into reliable capability. Its failures often came from practical problems that information alone could not solve. LLMs may help users understand and organize information, but they cannot validate experiments, manage hazards, or guarantee sound judgment. They can also mislead users by oversimplifying hard problems or producing confident but incorrect explanations. Information access and operational capability should not be treated as the same thing.

The second counterargument is case selection. Aum was not typical. It had unusual resources, educated personnel, international reach, and a leader willing to push the organization toward apocalyptic violence. Most violent non-state actors do not have that combination. That makes Aum useful as a stress test rather than a representative case. The case does not show that LLMs would meaningfully uplift every extremist, lone actor,

or violent organization. It shows how LLMs could affect one important actor type: a motivated group with resources and partial expertise, yet still struggling to turn ambition into execution.

The third limit is counterfactual uncertainty. This paper asks how LLMs could affect an Aum-like actor, but no historical case can prove that answer. Aum operated in a specific time and organizational environment. Modern LLMs would have changed some conditions, but they would not have removed Aum's internal dysfunction. The historical record also remains incomplete. Accounts rely on interviews, trial testimony, press reporting, and expert reconstruction, and those sources do not always align.

The fourth counterargument is that AI can strengthen defense. It can improve detection, intelligence analysis, response coordination, and preparedness. Model safeguards, platform monitoring, law enforcement, biosafety oversight, and public health systems all shape the risk environment. Aum's own history shows that external pressure mattered, as even imperfect law-enforcement attention disrupted timelines, altered decisions, and forced the group to react.

Threshold-lowering is not guaranteed success. It means a motivated actor may cross a practical line more easily than before. That line may involve better understanding, faster planning, improved coordination, or a higher chance of learning from failure. None of these changes ensures operational success. Together, however, they can change an actor's risk profile.

Implications for Frontier AI Labs

The Aum case gives frontier AI labs a practical lesson. CBRN safety evaluations should not stop at whether a model gives an obviously prohibited answer in a single exchange. That test still matters, but it misses much of the threshold-lowering problem. Aum's failures did not stem from a single missing fact. They appeared across planning, coordination, failure interpretation, expertise fit, and organizational learning.

Harmful capability uplift provides a stronger frame. The issue is whether an LLM helps a motivated user do more than they could have done without the model. Vaccaro et al. argue that safety evaluations should assess the human-AI system rather than isolated model outputs, because static benchmarks, human ratings, and red teaming often miss how users incorporate model assistance into sustained harmful workflows. That distinction fits the Aum case. A model could still help a user organize, persist, and learn across a longer project even when it refuses to provide clearly prohibited answers (Vaccaro et al., 2026).

This also helps avoid a safetywashing problem. Safety benchmark gains do not necessarily prove that a system has become safer in the ways that matter for misuse. Ren et al. warn that some safety benchmarks track general capability gains, which can make ordinary model improvements appear as safety progress. For CBRN risk, labs should therefore ask a harder question: does the system reduce harmful capability relative to its overall capability, or does it simply perform better on benchmarked refusals while still helping users across adjacent tasks? (Ren et al., 2024).

Planning and troubleshooting should count as safety-relevant capabilities. These functions create much of the legitimate value of LLMs, so labs cannot remove them from general-purpose systems without damaging beneficial use. The same functions can support harmful activity when paired with malicious intent. In an Aum-like case, refusing direct technical instructions may not eliminate risk if the model continues to support planning, review, and next-step selection. A model that helps a user preserve context, review failed assumptions, and choose a more coherent next step may contribute to uplift.

A misuse-chain approach would make evaluations more realistic. Aum did not move from ideology to harm in one step. It moved through motivation, resource gathering, technical exploration, failed attempts, adaptation, and execution. RAND's AI-CBRN research agenda supports this kind of pathway analysis, arguing that current evaluation efforts remain fragmented and need more dynamic, causal-chain-based approaches that account for AI tool coupling, decision support, and incident reporting (Hersman, Blackwelder, & Parachini, 2026). The 2026 International AI Safety Report reinforces the same point from a broader perspective on frontier AI. It notes that pre-deployment testing has become harder as models improve at recognizing evaluation settings and exploiting loopholes. At the same time, technical safeguards remain limited and work best when layered as part of a defense-in-depth approach (Bengio et al., 2026). Model Evaluation and Threat Research's (METR) review of frontier AI safety policies shows that leading labs already use related concepts, including capability thresholds, deployment mitigations, model weight security, evaluation frequency, and conditions for halting deployment or development (METR, 2025).

Current U.S. policy also recognizes the national security dimension of frontier-model evaluation. *America's AI Action Plan* identifies frontier AI systems as a potential source of novel national security risks, including risks related to cyberattacks and the development of CBRNE weapons. It recommends that the Center for AI Standards and Innovation (CAISI) at the Department of Commerce lead frontier-model evaluations for national security risks in partnership with frontier AI developers and agencies with relevant CBRNE and cyber expertise (The White House, 2025).

Capability thresholds should trigger governance action. Emerging regulation already points in this direction. The European Union's AI Act imposes additional obligations on providers of general-purpose AI models that pose systemic risk. That framework, by itself, does not solve the CBRN uplift problem. Still, it reinforces the central point that once a model materially increases user capability in a high-risk domain, the result should trigger governance action (European Union, 2024, Recital 112). The Frontier Model Forum's biosafety threshold brief identifies an emerging industry focus on whether a model significantly enables people with limited scientific expertise to obtain, produce, or deploy biological threats compared with other available resources. It also emphasizes comparative baselines, actor expertise, real-world feasibility, and cumulative evidence rather than a single test result (Frontier Model Forum, 2025). This perspective has direct implications for labs. If evaluations

show that a model materially improves a user's ability to navigate a harmful workflow, it should trigger mitigation decisions, monitoring requirements, deployment limits, external review, or release delays.

General red teaming is not enough for this problem. Red teams can reveal whether a model produces prohibited content, but they do not always show whether a human can use the model to become more capable over time. CBRN misuse evaluation needs domain-specific judgment from people who understand both AI systems and real-world CBRN pathways. A response may appear harmless to a general evaluator, yet still help a motivated user organize a dangerous project. Domain experts can better identify when planning support, terminology clarification, failure review, or adjacent technical help starts to function as capability uplift.

Labs should also watch adjacent requests. A harmful actor may not ask directly for CBRN assistance. The user may ask for ordinary forms of support that become meaningful only in context. Many of those requests have legitimate uses, so labs should not treat every technical or planning question as suspicious. The better approach is context-aware detection that looks for persistent efforts to overcome barriers. The goal is not to block ordinary learning. It is to identify patterns consistent with capability-seeking behavior.

Frontier labs should also strengthen external accountability. The Future of Life Institute's AI Safety Index found that only a minority of leading firms reported substantive testing for dangerous capabilities that pose large-scale risks, such as bioterrorism or cyberterrorism, and criticized gaps in external third-party evaluation and risk-assessment methodologies (Future of Life Institute, 2025). For CBRN uplift, labs should not rely only on internal confidence. They should support independent review, publish enough methodological detail to make claims meaningful, and create structured ways to share incident data without exposing sensitive user, proprietary, or law-enforcement information.

Finally, frontier labs should not treat this as a problem that any single company can solve on its own. Non-state actors can move across tools, providers, and jurisdictions. Chan et al. argue that inconsistent safeguards create opportunities for malicious users to seek out the least restrictive systems, a problem they describe as safety arbitrage. They also recommend shared deployment guidelines, information-sharing on malicious use, and emergency communication channels for AI-related crises involving non-state actors (Chan et al., 2026). A lab can evaluate cumulative uplift seriously and still lose ground if less-careful systems pose fewer barriers for the same user.

For frontier AI labs, the lesson is practical: planning, troubleshooting, expertise translation, and iterative learning should be evaluated as safety-relevant capabilities when they measurably increase user capability.

Policy and Safety Recommendations

The Aum case points toward a tiered-access model for CBRN-related AI assistance. Public models should not treat every CBRN topic as prohibited. Historical analysis, public health, emergency preparedness, arms-control research, and safety education all have legitimate value. The stronger boundary should fall around capability-enabling assistance. Public systems should allow discussion and analysis without becoming project assistants for unvetted users pursuing CBRN capability.

This approach fits the dual-use problem identified in the Aum case. Aum's failures were evident in planning, judgment, coordination, and learning. Those are ordinary areas where LLMs are useful. The same features that help a responder organize a preparedness plan can help a motivated actor sustain momentum throughout a harmful project. The policy challenge is not CBRN information by itself. Sustained assistance can move an unvetted user closer to practical capability.

Tiered access is not a novel governance instinct. Current U.S. biosecurity policy reflects a similar risk-based logic in the biological domain. Executive Order 14292 frames dangerous gain-of-function research as a public health, economic, and national security concern, reinforcing the principle that some biological workflows require stronger oversight than ordinary scientific discussion (Executive Office of the President, 2025).

In other dual-use domains, governments often preserve broad beneficial use while placing stronger controls around enabling capabilities. Export controls on advanced semiconductors restrict access to key inputs for military-relevant AI and supercomputing without prohibiting AI research as a field (Federal Register, 2022). Nuclear governance draws a sharper line between general nuclear knowledge and Restricted Data or Critical Nuclear Weapon Design Information, where access depends on formal classification and need-to-know controls (U.S. Department of Energy, 2011). Global Positioning System (GPS) illustrates a less restrictive version of the same pattern: public access expanded after the United States ended Selective Availability, which had intentionally degraded civilian accuracy for security reasons (GPS.gov, n.d.). These examples are not perfect analogies to LLMs, but they support the same core principle: access can be structured by risk without suppressing the entire domain.

The central policy principle is risk-matched access. More sensitive workflows should move to secure-access environments, accessible only to vetted users through authorized organizations.

Model evaluations should reflect this structure. A model that passes single-turn refusal tests may still create uplift. The relevant test is whether users become more capable with the model than without it. Aum shows why: partially capable actors may gain more from assistance that organizes knowledge and improves decision-making than either complete novices or true specialists. If a model materially improves an actor's ability to navigate a harmful workflow, labs should treat the result as a deployment concern rather than a benchmark detail. Existing U.S. guidance already points in this direction. NIST's Generative AI Profile treats generative AI risk management as a lifecycle problem rather than a narrow question of whether one output violates a rule (Autio et al., 2024).

Memory, file uploads, and agentic tools make this problem sharper. These features allow work to continue across time. They can help a legitimate team maintain continuity, but they can also turn scattered assistance into a sustained learning cycle. Public systems should apply tighter controls when CBRN-relevant content appears alongside repeated attempts to organize, troubleshoot, or advance a project. In secure environments, those same features may be appropriate, but only because identity, purpose, logging, and oversight reduce the risk of unchecked capability-building.

No single lab can make tiered access work on its own. A motivated actor can move across providers, open-weight models, commercial tools, and jurisdictions. Frontier labs should coordinate on evaluation standards, incident reporting, and escalation channels. Government and CBRN domain experts should help shape those standards because general AI evaluators may miss when ordinary assistance begins to function as practical uplift.

The goal is not to suppress legitimate CBRN knowledge. It is to keep public models from becoming capability-building infrastructure for unvetted users. Tiered access gives frontier labs a practical way to preserve beneficial use while placing stronger controls around the assistance most likely to lower the threshold for harm.

Conclusion

Aum Shinrikyo shows why CBRN capability cannot be reduced to intent or resources. The group had both, along with educated personnel, money, facilities, international reach, and a willingness to pursue mass-casualty violence. Even with those advantages, Aum struggled to turn ambition into reliable execution. Its history exposes the gap between wanting to cause catastrophic harm and being able to do so effectively.

That gap is where LLMs enter the risk analysis. They do not need to remove every constraint or provide direct operational guidance to change risk. Their relevance lies in how they help users turn scattered knowledge into more organized action. Those functions also give LLMs enormous legitimate value. In the hands of a motivated actor, however, the same functions can lower the threshold between violent intent and practical capability.

Threshold-lowering risk should be evaluated separately from ceiling-raising risk. Ceiling-raising focuses on advanced actors using AI to push the upper limits of what is technically possible. Threshold-lowering focuses on motivated actors with some resources and partial expertise who still struggle to become competent enough to matter. Aum belongs in that second category. The case does not show that LLMs would have guaranteed success. It shows that marginal improvements in planning, troubleshooting, expertise translation, coordination, and learning could have mattered at several points where Aum historically failed.

Aum's failures should not reassure us too much. The group failed often, but failure did not make it harmless. It adapted, persisted, shifted pathways, and ultimately caused serious harm. That is the AI safety lesson. Frontier LLMs will not make every bad actor successful. They may help some motivated actors fail less often, learn more quickly, and cross thresholds that previously kept them below the line of credible harm.

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